



ITI Cmos Analog IC Design
Lab 07
OTA

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Section 7 online

Part 1 — OTA Design (gm/ID Methodology)

Target Specifications

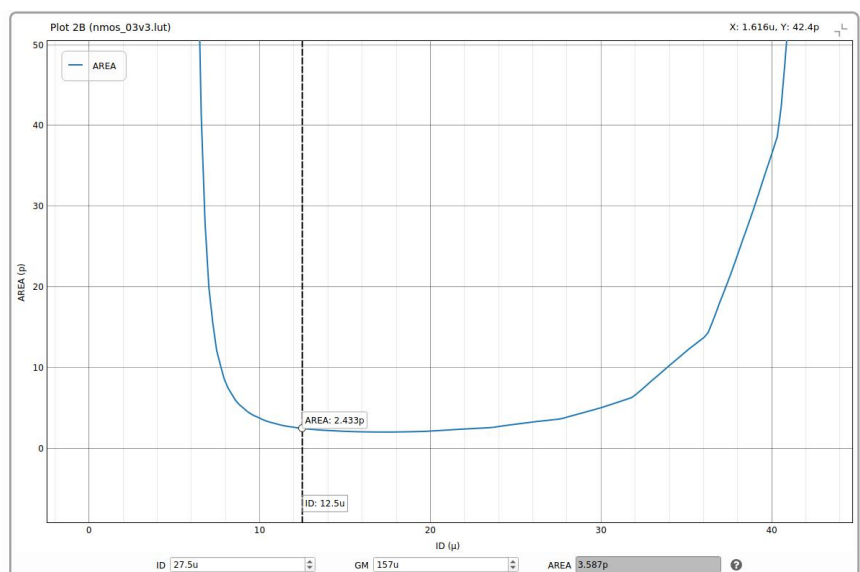
Parameter	Requirement
Supply voltage, VDD	1.8 V
Load capacitance, CL	5 pF
Open-loop DC gain, Av0	≥ 34 dB (≥ 50.1 V/V)
CMRR @ DC	≥ 74 dB
Phase margin, PM	$\geq 70^\circ$
Reference (seed) current	10 μ A
CM input range — low	≤ 1.2 V
CM input range — high	≥ 1.5 V
GBW	≥ 5 MHz
Current consumption / Area	Minimize

The required gain ($34 \text{ dB} \approx 50 \text{ V/V}$) is modest, so a single-stage amplifier will be sufficient since the CMIR spec is close to VDD rail so NMOS input pair will be used in the design with pmos current mirror load + nmos tail current mirror.

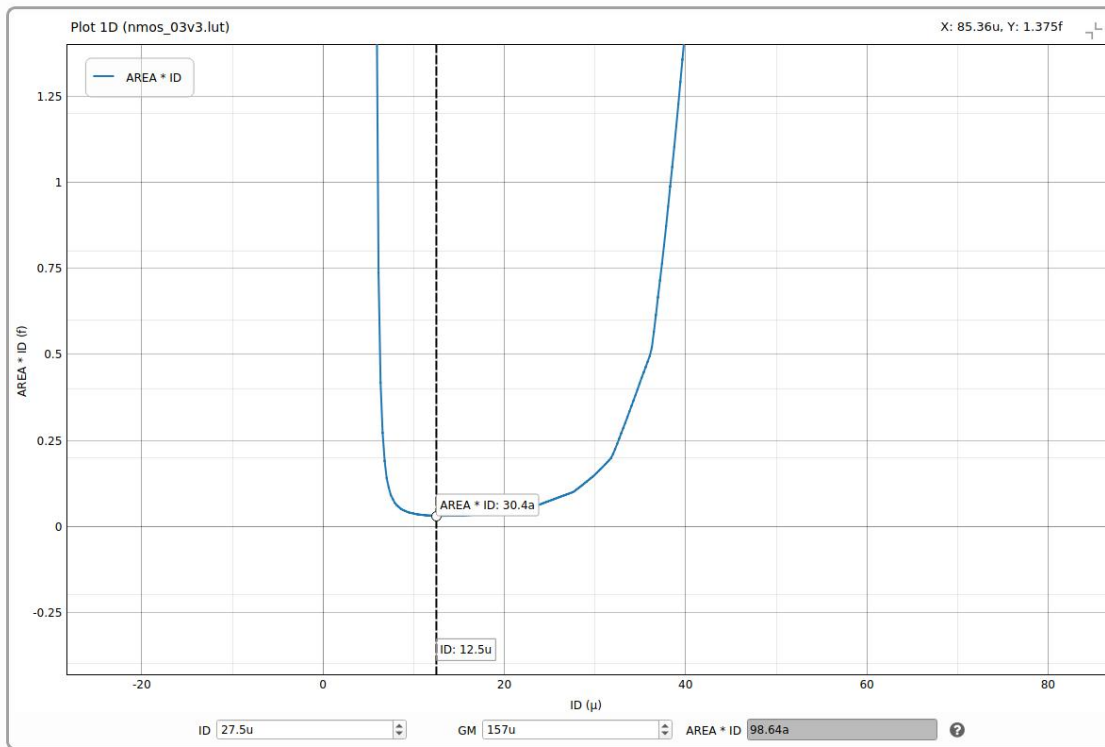
1.2 input pair sizing

- From $GBW = gm * 2 * \pi * CL \implies gm = 160 \mu$
assume $ro_{1,2} = ro_{3,4}$ so $gm/gds = 100$
- Sweeping id and using gm as knob and gm/gds as a condition.

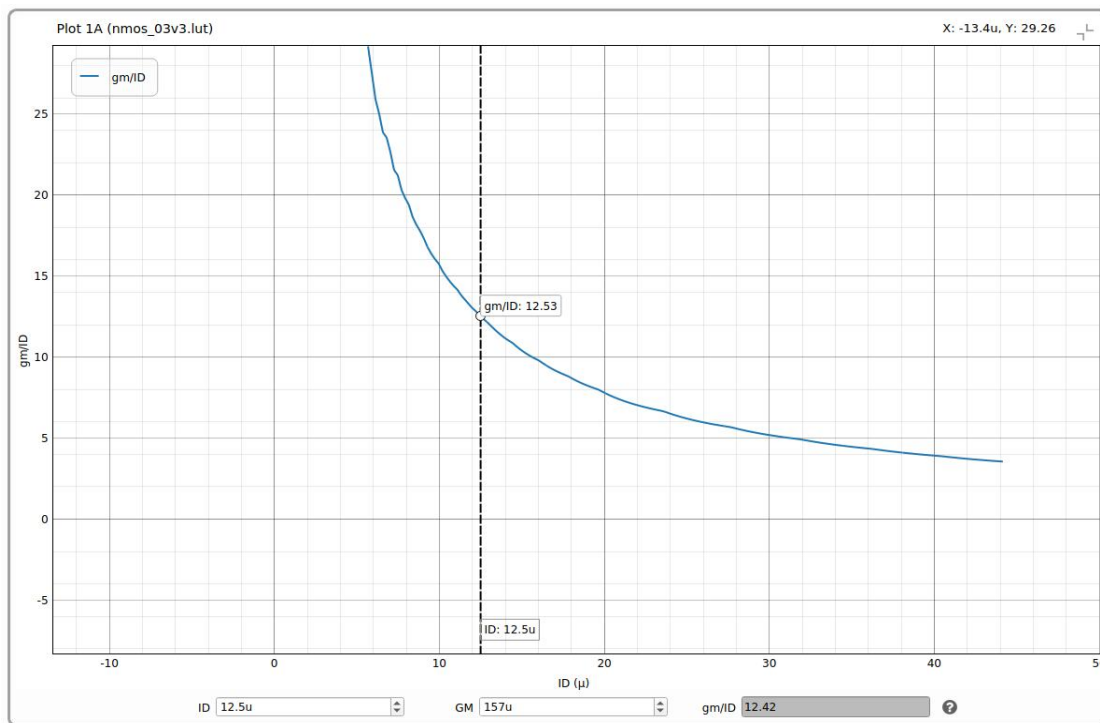
Area vs id

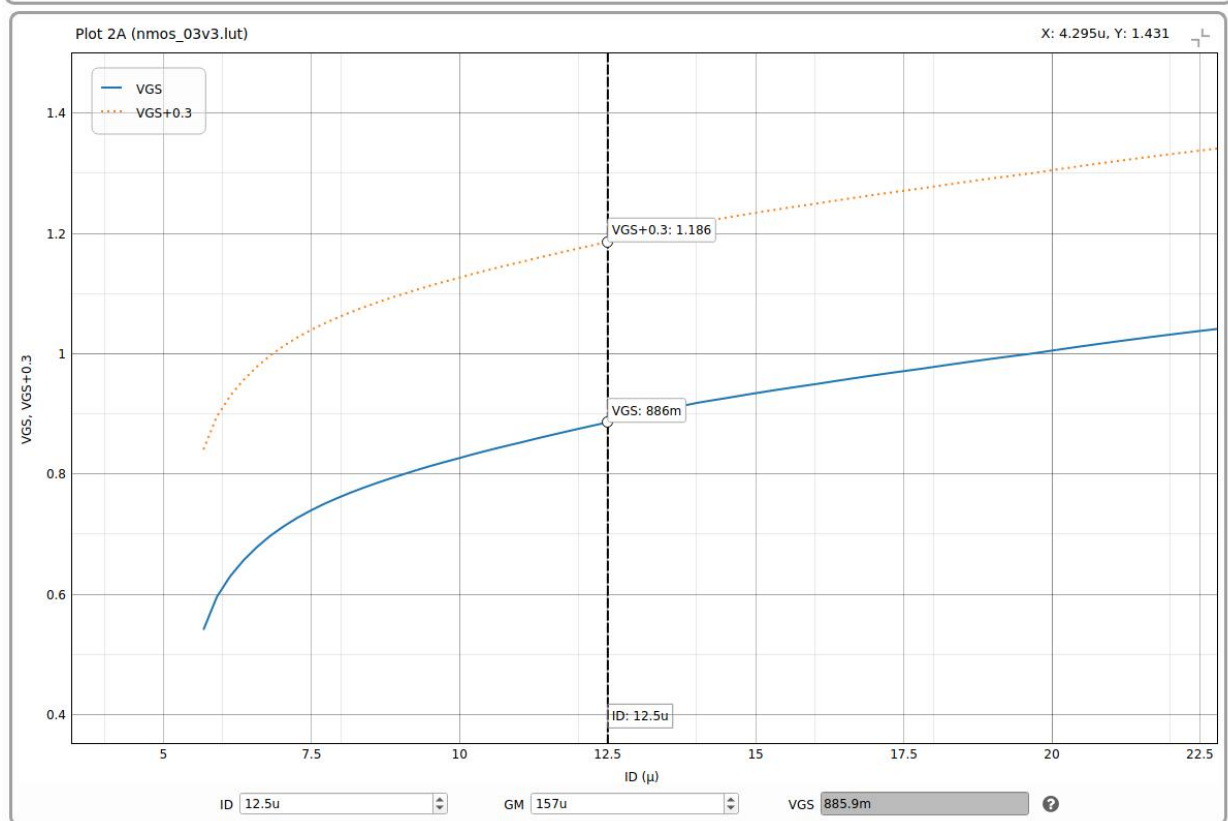
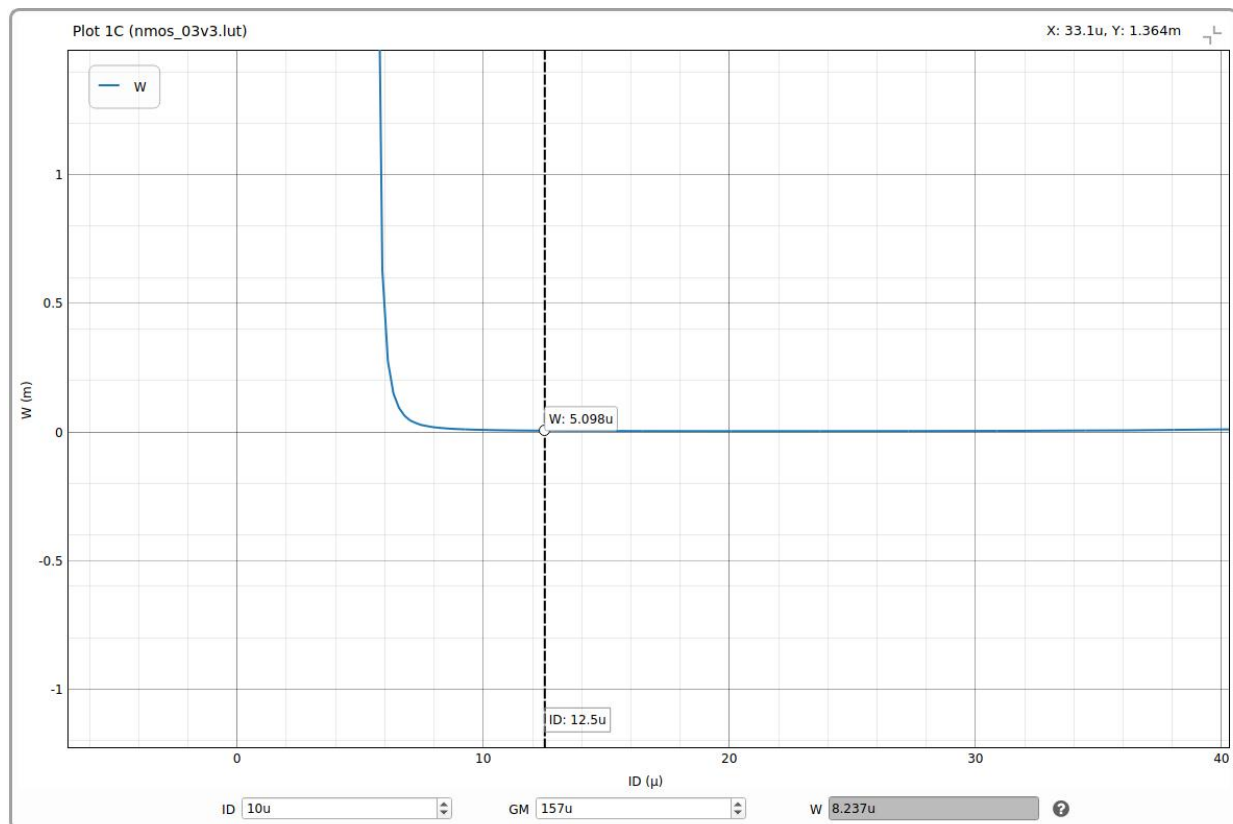


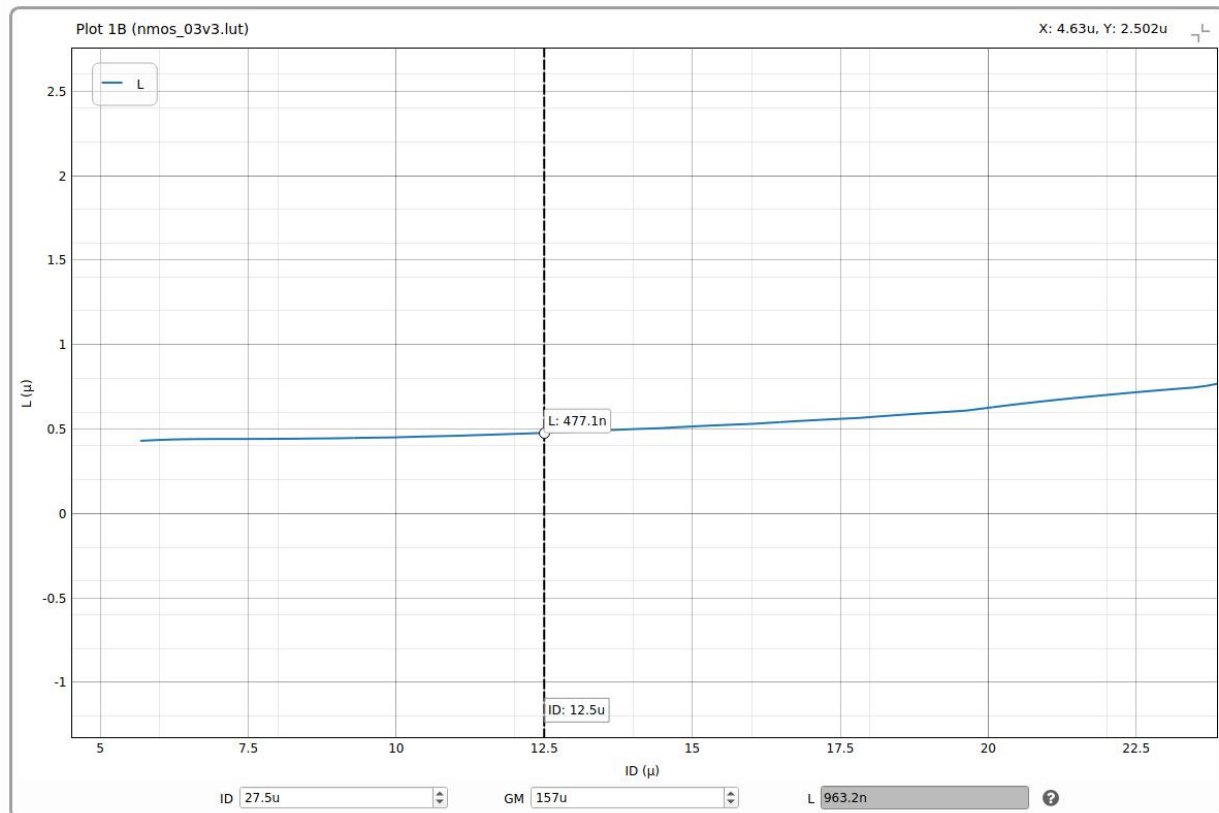
Area*id vs id



Gm/id vs id







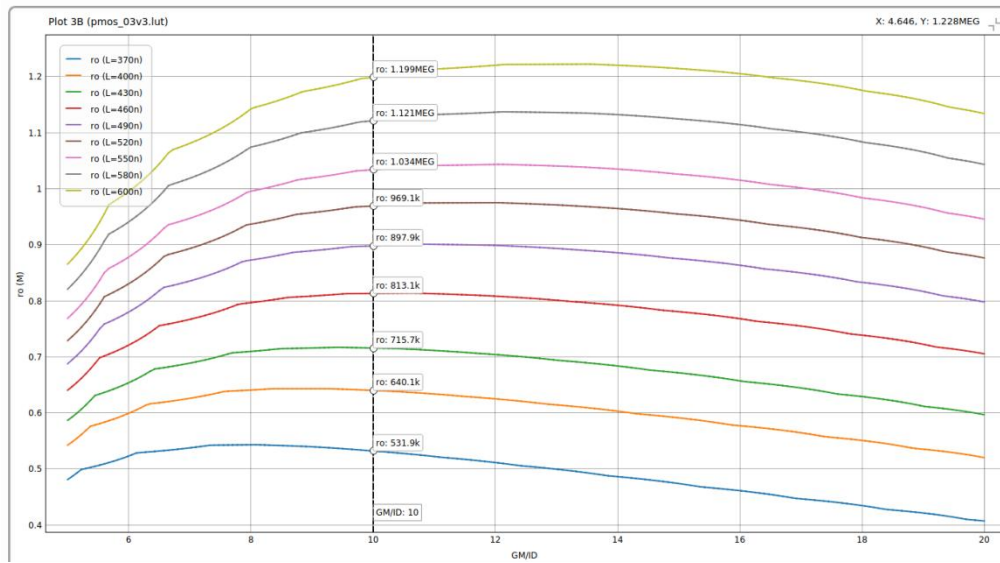
- from graphs to minimize area
we will use **id = 12.5** and **gm/id = 12.5**
- From graph $V_{incm,min} = V_{GS,in} + V_{DS,tail} = 1.186 > 1.2 \text{ V}$
- Also $L = 477\text{n}$ and $W = 5.1 \text{ u}$

1.3 pmos Active load Sizing

- Assume $Gm/id = 10$
- From $gm1(ro1 || ro3) = 100 \implies ro3 = 630 \text{ K}$

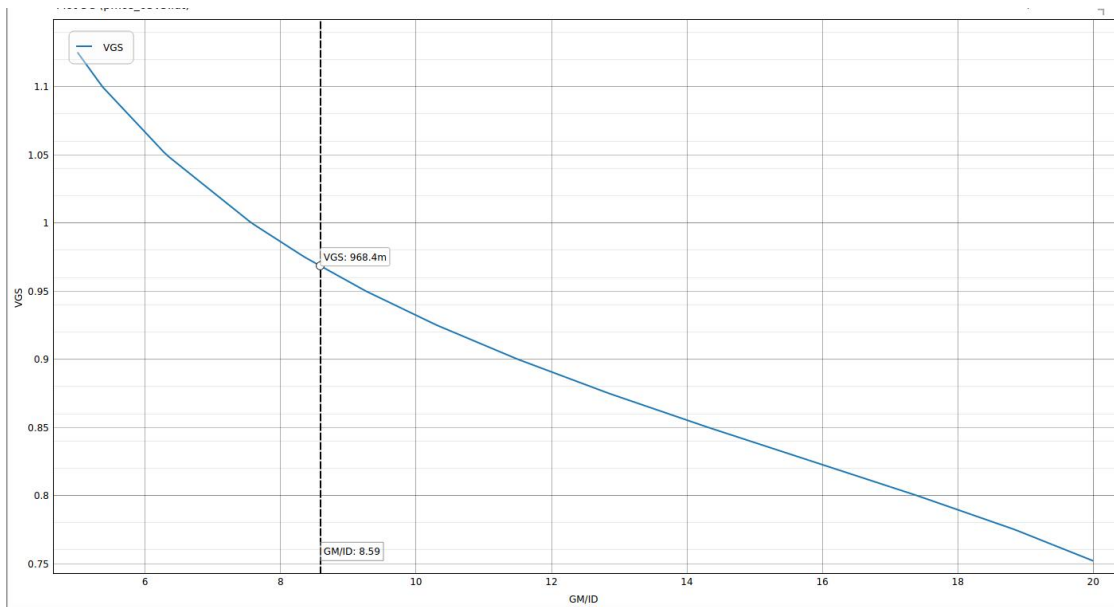
ID	12.5u	?
gm/ID	5:20	?
L	400n:30n:600n	?
VDS	0.5	?
VSB	0	?
Stack	1	?
Results:		

- Plot r_o vs g_m/i_d

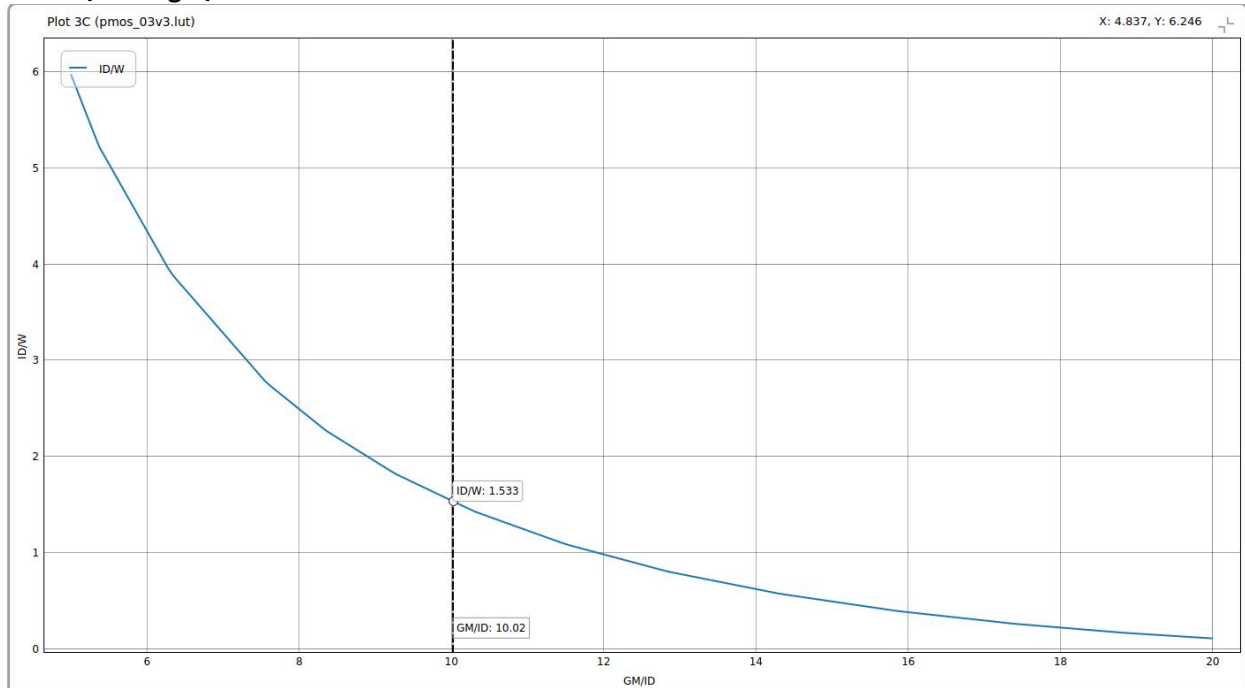


From graph at $g_m/i_d = 10$ and $r_o = 640k$ at $L = 400n$

- $V_{incm_high} = v_{gs1} - v_{dsat1} - v_{gs3} + v_{dd} > 1.6$
 $V_{gs3_max} = 966$ mV from graph below min $g_m/i_d = 8.59$ so $g_m/i_d = 10$ is good



● I_d / w vs gm/id



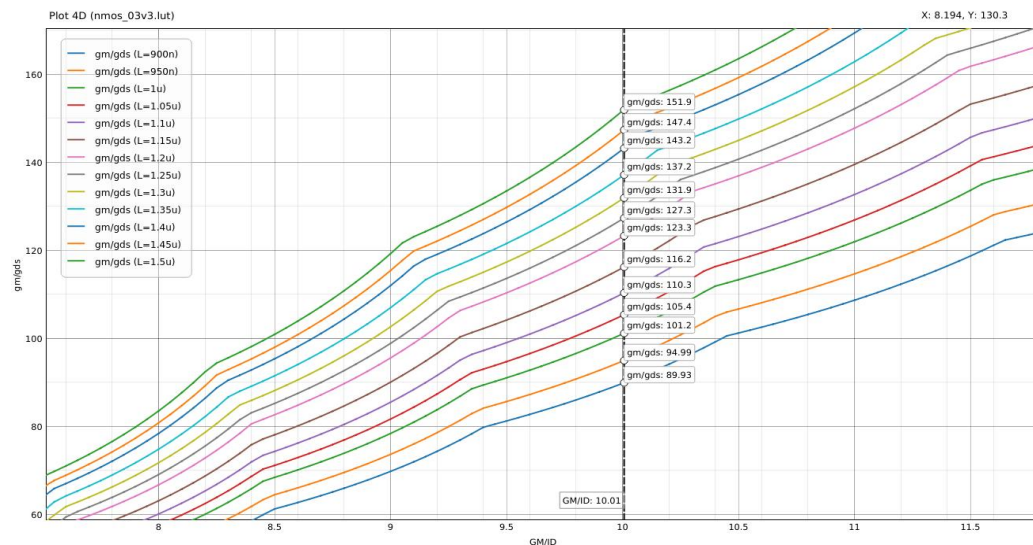
● $W = id / (id / w) = 8.1168 \mu$

1.3 Nmos tail current mirror

- From $v_{icm_low} = V_{GS1,2} + V_{DSsat5} < 1.2 \implies v_{dsatmax} = 0.309$ maximum $gm/id = 5$
- Assume $gm/id = 10$ current = 25μ so so $gm = 250\mu$
- $CMRR = 2 * gm_{3,4} * R_{SS} = 74 \implies R_{SS} = 405K$
- $Gm/gds = 100.1$

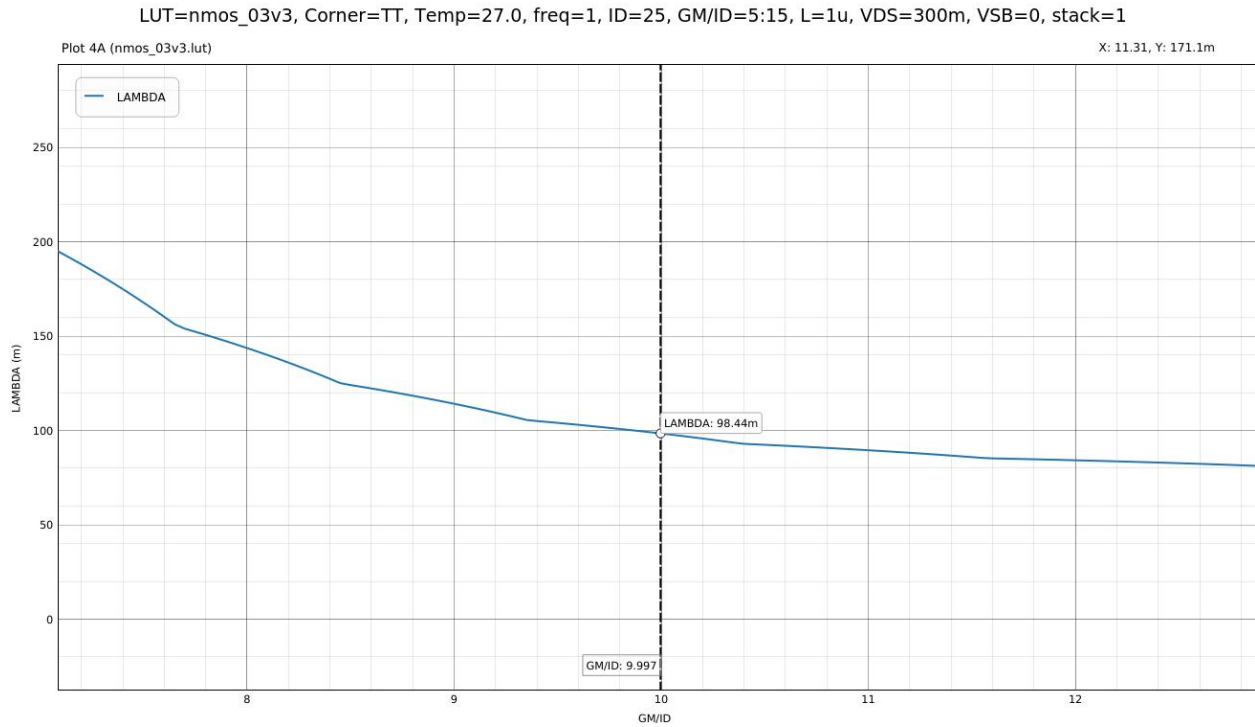
GM /gds vs gm/id

corner=TT, Temp=27.0, freq=1, ID=25, GM/ID=5:15, L=1.05u/1.15u/1.1u/1.25u/1.2u/1.35u/1.3u/1.45u/1.4u/1.5u/1u/900n/950n, VDS=300

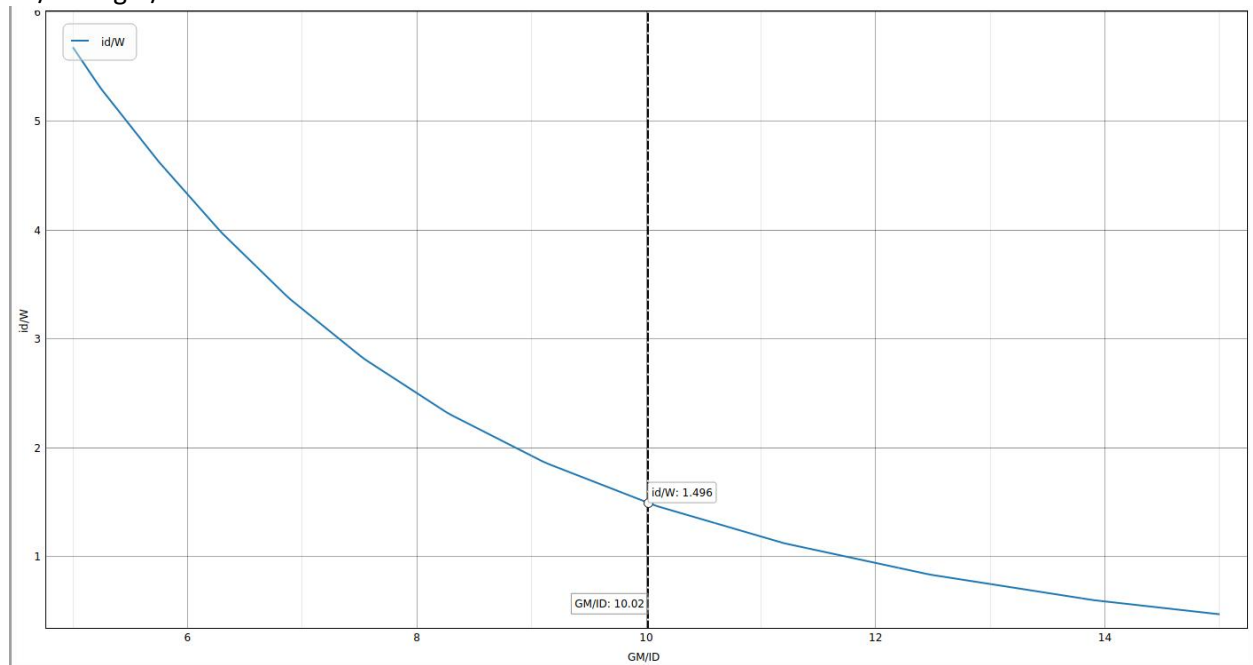


From graph at $L = 1\mu$ $gm/gds = 101.1$

- Lambda VS gm/id at L = 1u



- Id /W vs gm/id



- At gm / id = 10, ID / W = 2.142

Tail current mirror will be 5 of unit size 5u current

for unit size 5u , $W = id / (id/W) = 2.33u$

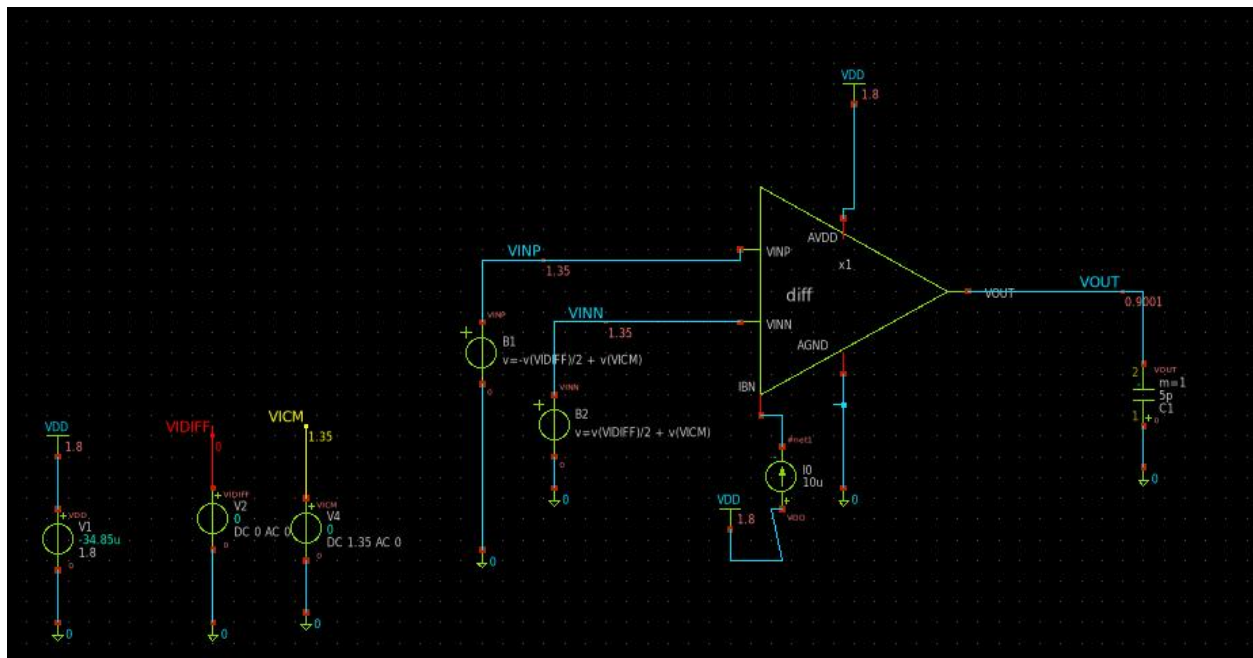
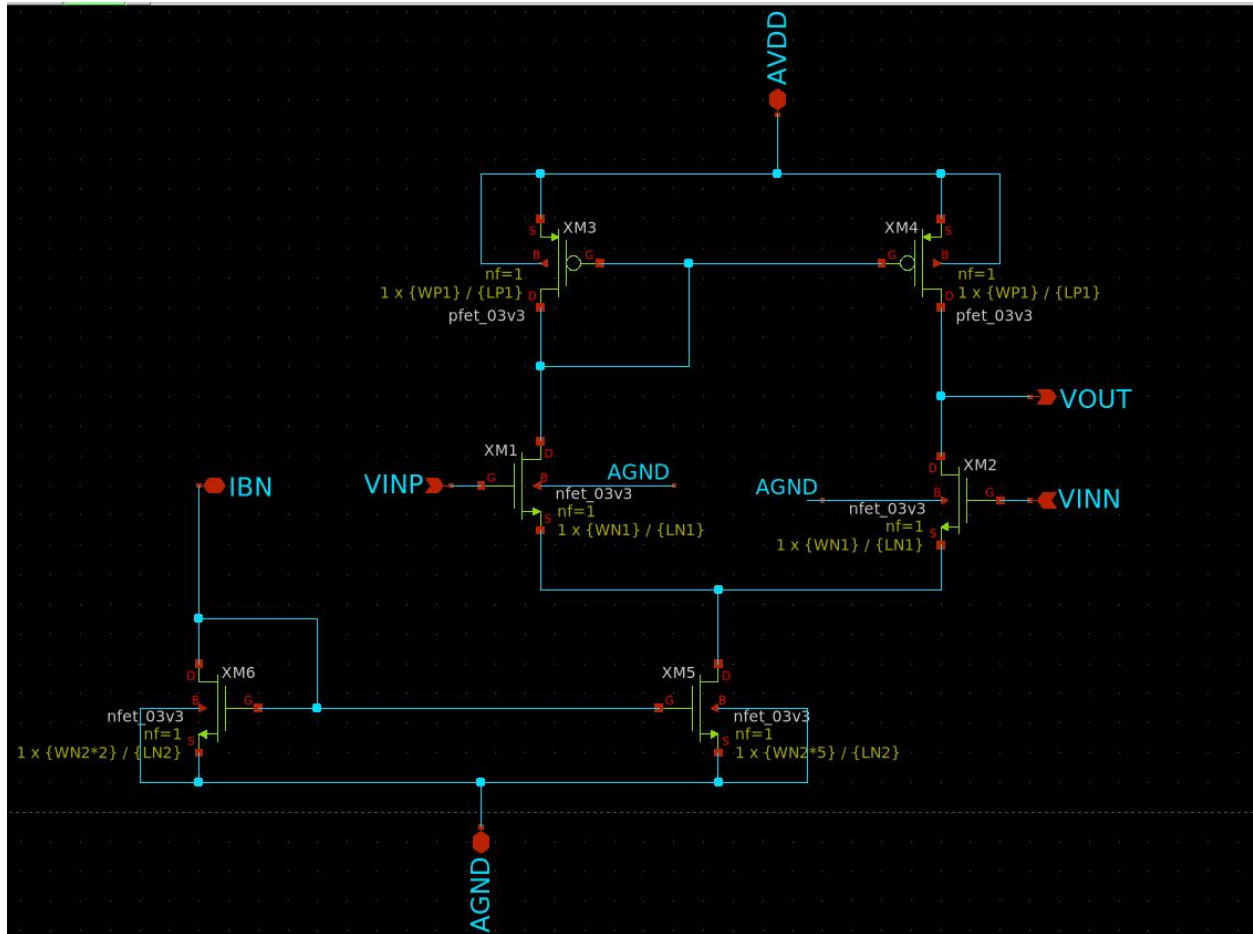
So for tail current mirror $W = 5 * 2.33u = 11.67 u$

$\text{Sqrt}(1+1/(5/2)*id_{mis}/id*100) = 1.7\%$ this point

Device	Role	ID (μA)	gm ($\mu\text{A/V}$)	gm/ID (S/A)	W (μm)	L (m)	VGS (m)	Vov=VGS-VTH (mV)	Vdsat (mV)	V*=2/(gm/ID) (mV)	Power * area ($\mu\text{A} \cdot \mu\text{m}^2$)
M1,2	NMOS input +	12.5	157	≈ 12.5	5.1	477n	891	85.98	138.5	160	30.4
M3,4	PMOS mirror	12.5	125	10	8.11	400n	941	156.3	177.4	200	40.5
M5	NMOS tail source	25	250	10	11.64	1u	853	149	172	200	291

Total power * ID = $2(30.4)+2*(40.5)+291= 432.8$ ($\mu\text{A} \cdot \mu\text{m}^2$)

Part 2 — Open-Loop OTA Simulation



2.1 DC Operating Point

```
--- MOSFET operating points ---
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm6.m0      m.x1.xm5.m0      m.x1.xm2.m0
model       nfet_03v3.9      nfet_03v3.13     nfet_03v3.8
id          1e-05            2.4852e-05       1.2426e-05
vgs         0.807505         0.807505         0.921616
vds         0.807503         0.428378         0.471667
vth         0.687818         0.688438         0.84375
vdsat       0.152272         0.152471         0.133995
gm          0.000111965      0.000279487      0.000159295
gds         3.52065e-07      1.25607e-06      1.53868e-06
gmbs        4.28439e-05      0.000106985      4.52704e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm1.m0      m.x1.xm4.m0      m.x1.xm3.m0
model       nfet_03v3.8      pfet_03v3.8      pfet_03v3.8
id          1.2426e-05       1.2426e-05       1.2426e-05
vgs         0.921616         0.899944         0.899944
vds         0.471667         0.899942         0.899942
vth         0.84375          0.773511         0.773511
vdsat       0.133995         0.155909         0.155909
gm          0.000159295      0.000140206      0.000140206
gds         1.53868e-06      1.37988e-06      1.37988e-06
gmbs        4.52704e-05       5.59728e-05      5.59728e-05

=== Bias Points ===
M1: gm/id = 12.8195 S/A, Vstar = 0.156012 V
M2: gm/id = 12.8195 S/A, Vstar = 0.156012 V
M3: gm/id = 11.2833 S/A, Vstar = 0.177253 V
M4: gm/id = 11.2833 S/A, Vstar = 0.177253 V
M5: gm/id = 11.246 S/A, Vstar = 0.17784 V
M6: gm/id = 11.1965 S/A, Vstar = 0.178628 V
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

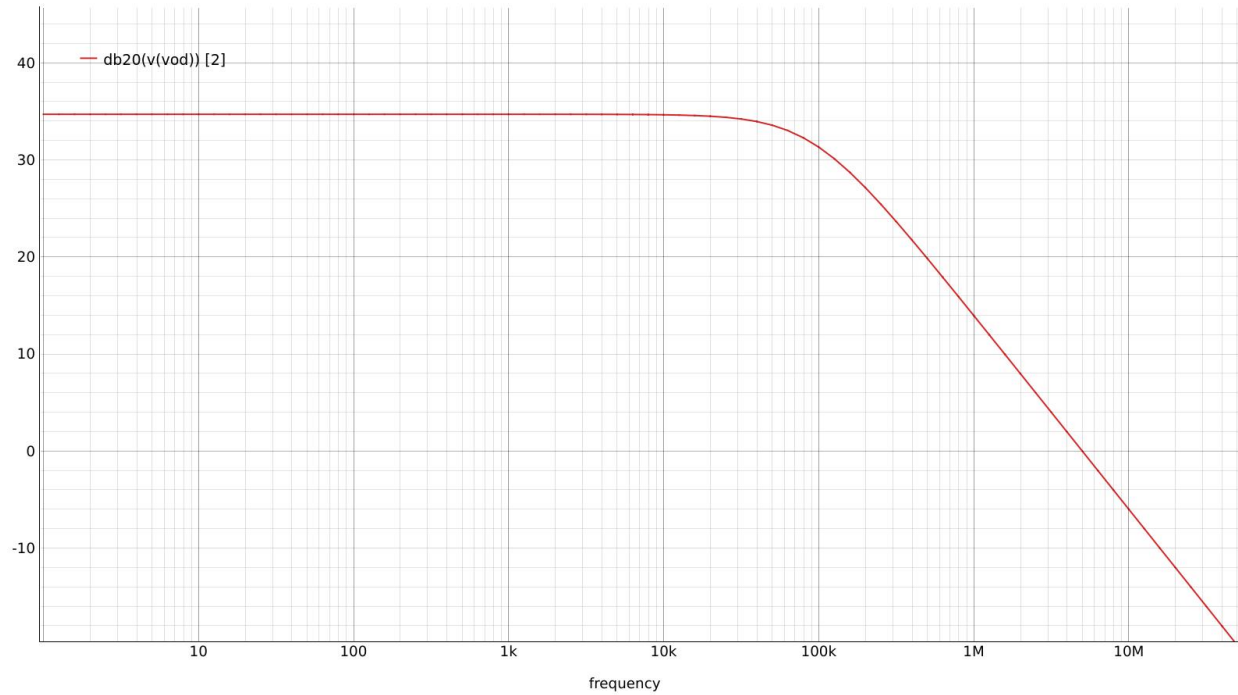
Is the current (and gm) in the input pair exactly equal?

Yes, exactly equal and also there is no mismatch since we disabled it in the code.

What is DC voltage at VOUT? Why?

Output voltage is 0.901 V is the output voltage when input differential is zero and equal V_F (mirror point) equal to $v_{dd} - v_{gs4} = 1.8 - 0.9 = 0.9$ V

2.2 Differential Small-Signal



Av Diff: 34.7057 dB, BW: 92183.6 Hz
UGF: 4.99921E+06 Hz
GBW: 5.01118E+06 Hz

Hand-calculation :

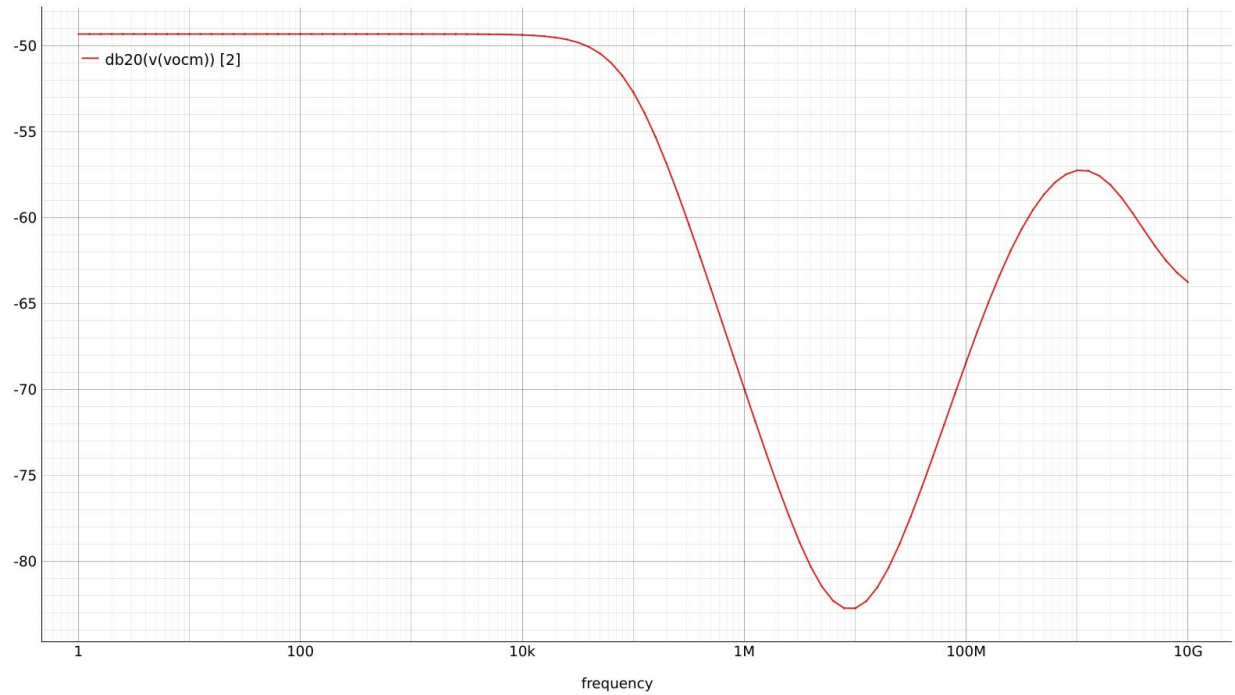
$$A_{v,diff} = g_{m1} \cdot (r_{o1} \parallel r_{o4}) = 54.6091 \text{ (34 dB)}$$

$$f_{p,dom} = 1 / (2\pi \cdot (r_{o2} \parallel r_{o4}) \cdot C_L) = 92.7 \text{ KHz}$$

$$GBW = g_{m1} / (2\pi \cdot C_L) = 5.0611 \text{ MHz}$$

Quantity	Hand calculation	Simulation	ERROR %
Av,diff (dB)	54.6 (34 dB)	45.36 (34.7 dB)	0.44 %
fp,dom (Hz)	92.7 KHz	92.1 KHz	0.65%
GBW (Hz)	5.0611 MHz	5.0111 MHz	0.99 %

2.3 Common-Mode Small-Signal



```

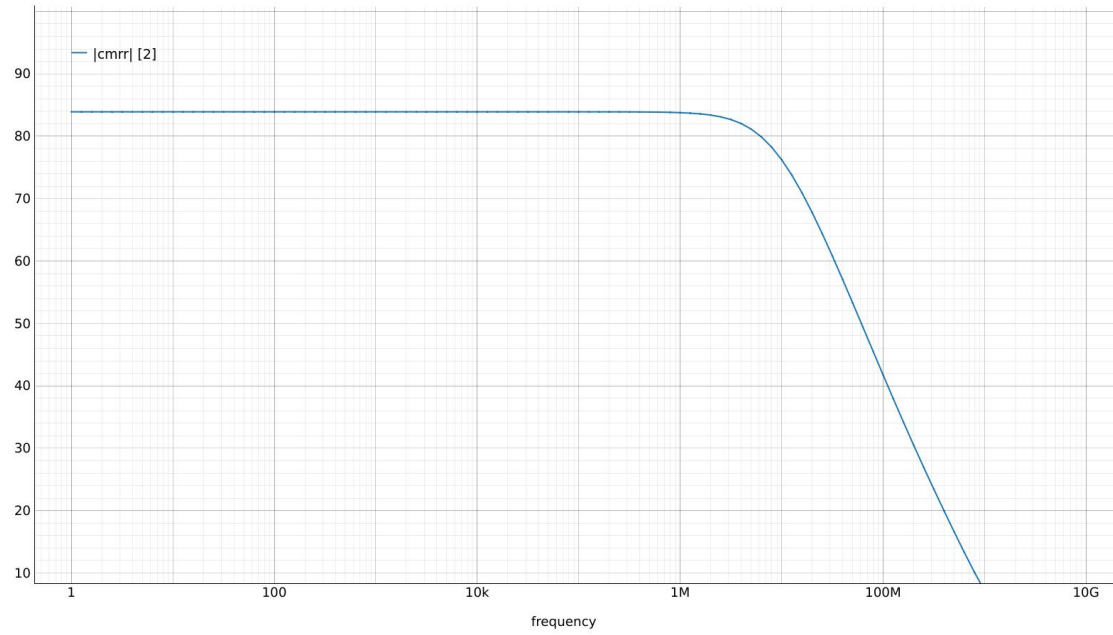
av_cm_db      = -4.92050e+01
av_cm_mag     = 3.46537e-03
bw_cm        = 9.22077e+04
  
```

Hand-calculation :

$$A_{v,cm} = 1 / (2 \cdot g_{m3} \cdot R_{ss}) = 4.479 \text{ m } (-47 \text{ dB})$$

Quantity	Hand calculation	Simulation	ERROR
$A_{v,cm}$ (dB) @ DC	4.479 m (-47 dB)	3.465 m (-49.2)	28.69 %

2.4 CMRR



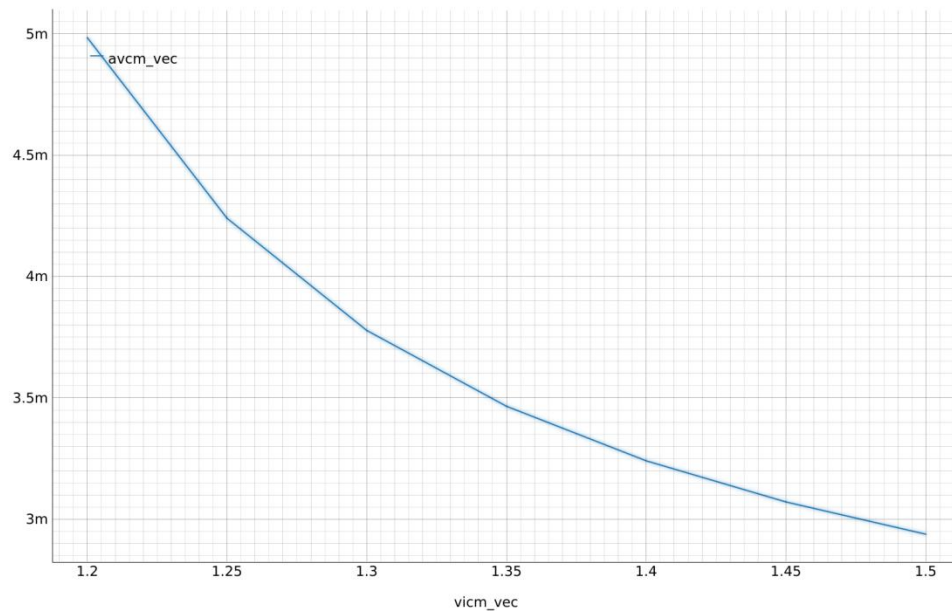
CMRR: 83.9107 dB

Hand-calculation :

$$CMRR = gm1 \cdot (ro1 \parallel ro4) \cdot (2 \cdot gm3 \cdot Rss) = 79.958 \text{ dB}$$

Quantity	Hand calculation	Simulation	ERROR
CMRR @ DC (dB)	79.958 dB	83.91 dB	4.7%

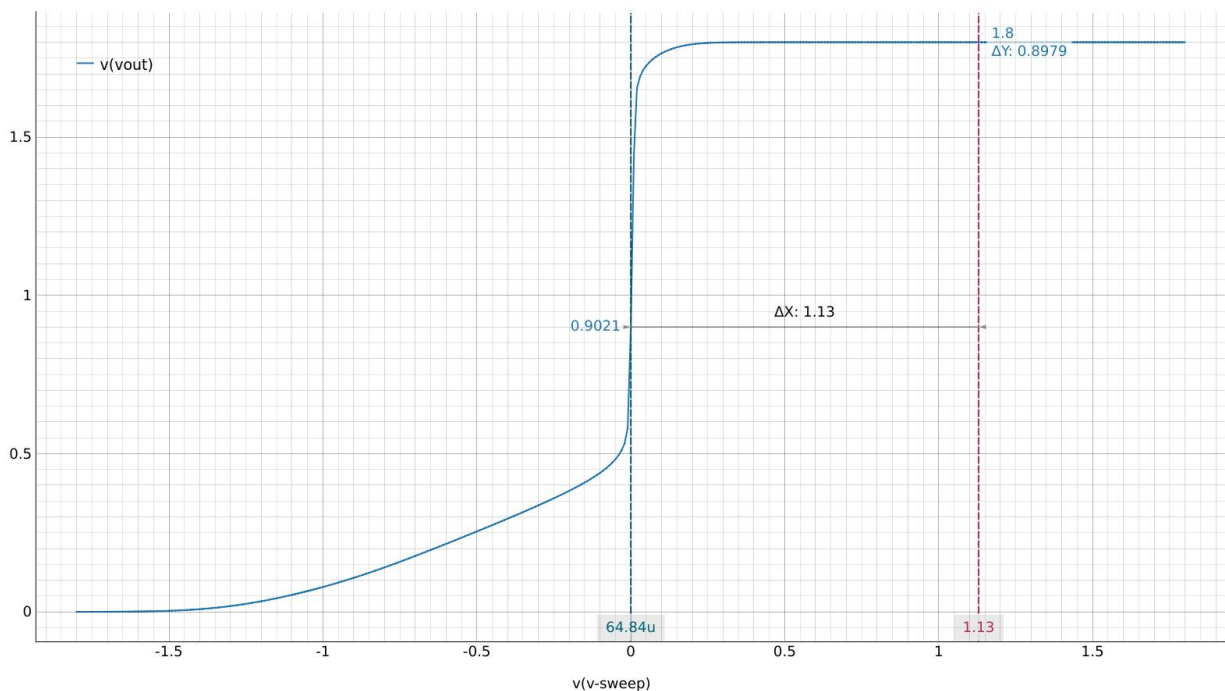
2.5 Common-Mode Gain vs. VICM (CMIR Sweep)



Justify the results. Does Avcm increase or decrease vs VICM? Why?

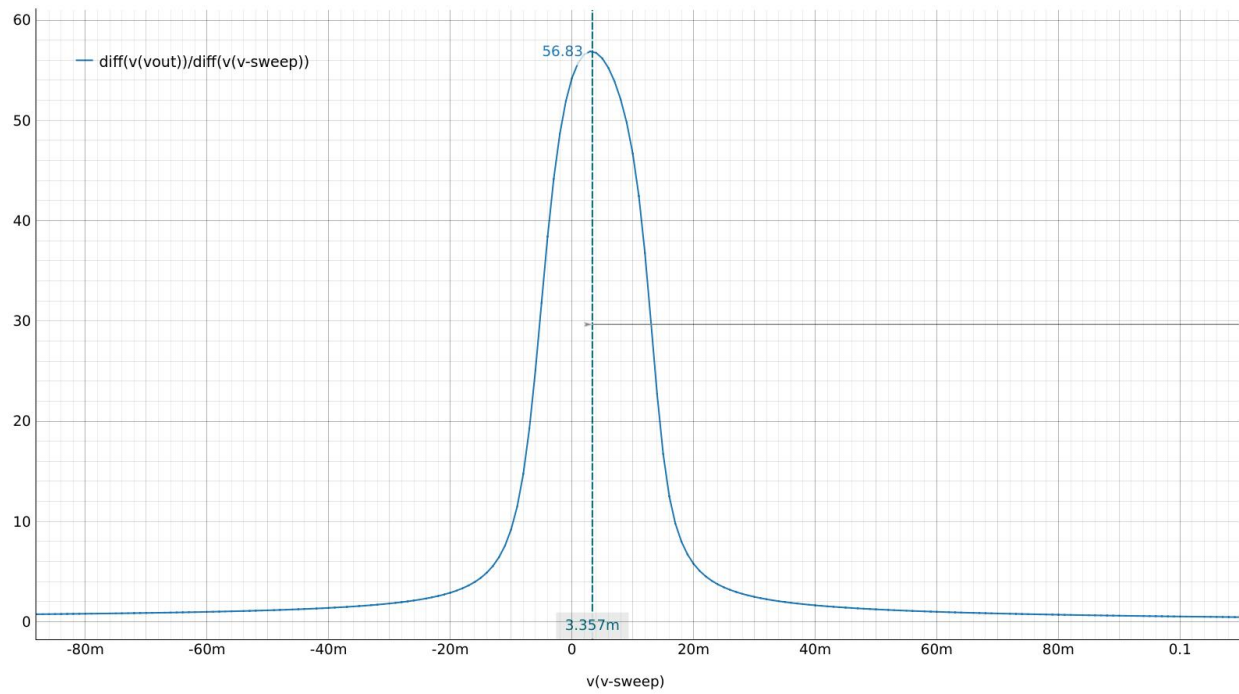
Common mode output gain is decreasing with increasing vicm value, the gain is determined by $(1 / (2 \cdot g_{m3} \cdot R_{ss}))$, when increase vicm the vds of tail current source increase meaning it goes deeper in saturation region which increases the R_{ss} of it that will reduce the common mode gain.

2.6 Differential Large-Signal



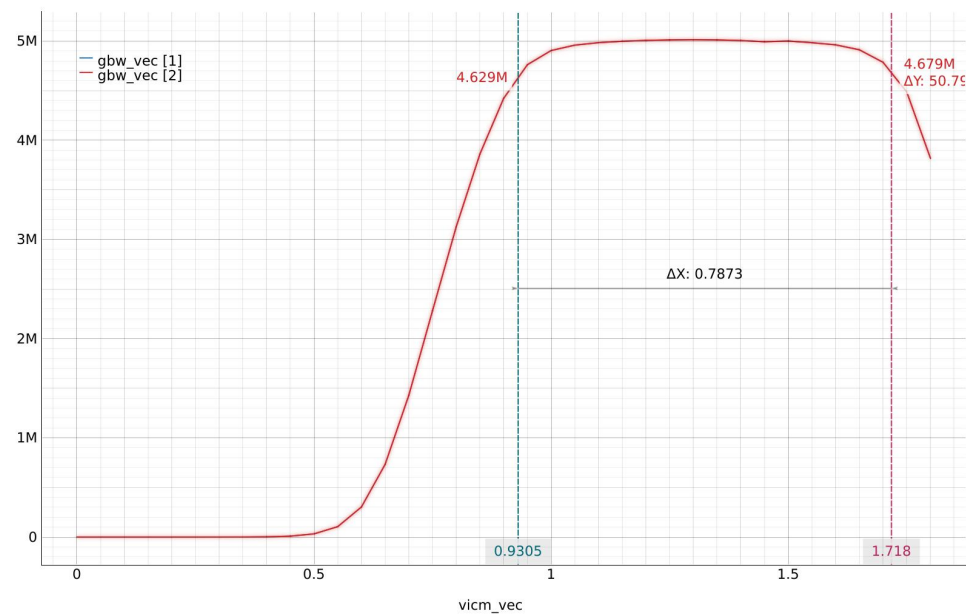
Value at vind = 0 is vout = 0.902 equal to Vmirror point $V_{DD} - V_{GS3} = 0.901$

Derivative of Vout vs Vin



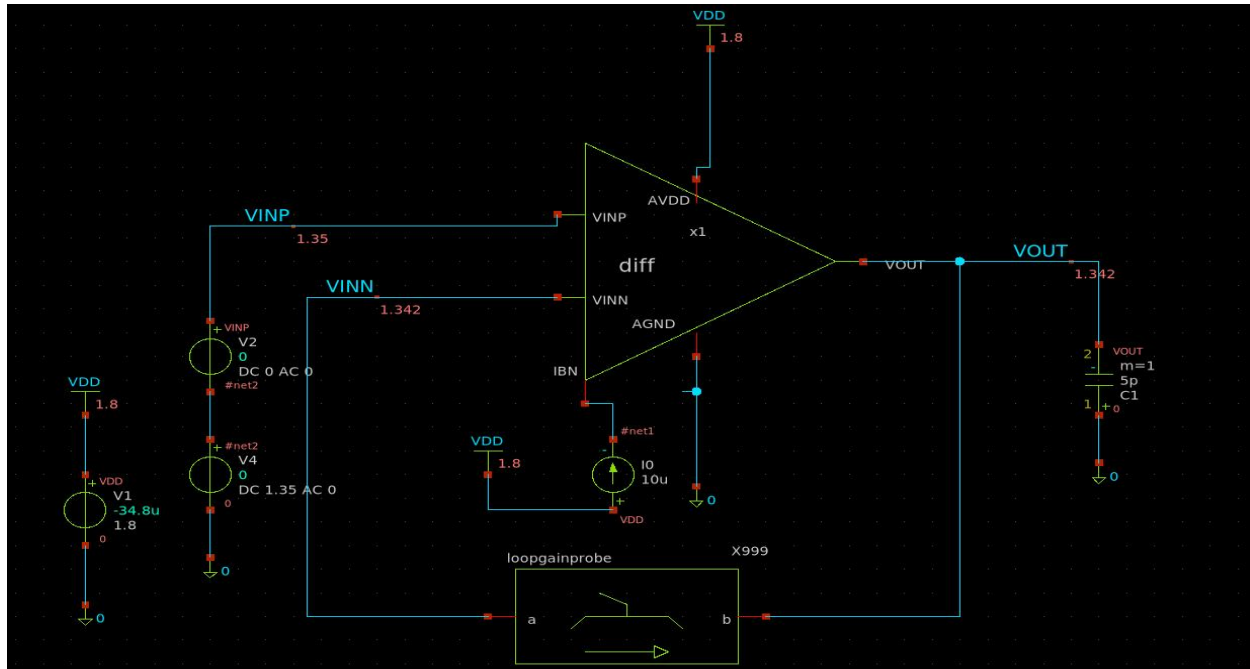
Peak value is 56 which is approximately equal to the dc gain when $v_{id} = 0$ that $\frac{dV_{out}}{dV_{in}} = A_{Vd}$

2.7 GBW vs. VICM (CM Input Range)



Quantity	Target	Extracted (90% of GBW)
CMIR low (V)	≤ 1.2	0.9305
CMIR high (V)	≥ 1.5	1.718

Part 3 — Closed-Loop OTA Simulation



3.1 Unity-Gain Buffer DC Operating Point

```

--- Model Operating Points ---
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm6.m0      m.x1.xm5.m0      m.x1.xm2.m0
model       nfet_03v3.10    nfet_03v3.14    nfet_03v3.8
id          1e-05           2.49272e-05     1.22266e-05
vgs         0.824766        0.824766        0.920866
vds         0.824764        0.422166        0.920863
vth         0.678463        0.678805        0.841037
vdsat       0.165333        0.165637        0.135239
gm          0.00010107      0.000252768     0.000154645
gds         2.60279e-07      9.72757e-07     1.06963e-06
gmbs        3.92562e-05      9.82247e-05     4.40828e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm1.m0      m.x1.xm4.m0      m.x1.xm3.m0
model       nfet_03v3.8      pfet_03v3.8      pfet_03v3.8
id          1.27005e-05      1.22266e-05     1.27005e-05
vgs         0.927827        0.940385        0.940385
vds         0.437438        0.456958        0.940383
vth         0.841827        0.786733        0.78604
vdsat       0.139353        0.171464        0.171986
gm          0.000157423      0.000123307     0.000127848
gds         1.65989e-06      1.27892e-06     8.02884e-07
gmbs        4.48742e-05      5.36484e-05     5.55768e-05

=== Bias Points ===
M1: gm/id = 12.395 S/A, Vstar = 0.161355 V
M2: gm/id = 12.6482 S/A, Vstar = 0.158125 V
M3: gm/id = 10.0664 S/A, Vstar = 0.198681 V
M4: gm/id = 10.0852 S/A, Vstar = 0.198311 V
M5: gm/id = 10.1402 S/A, Vstar = 0.197234 V
M6: gm/id = 10.107 S/A, Vstar = 0.197883 V

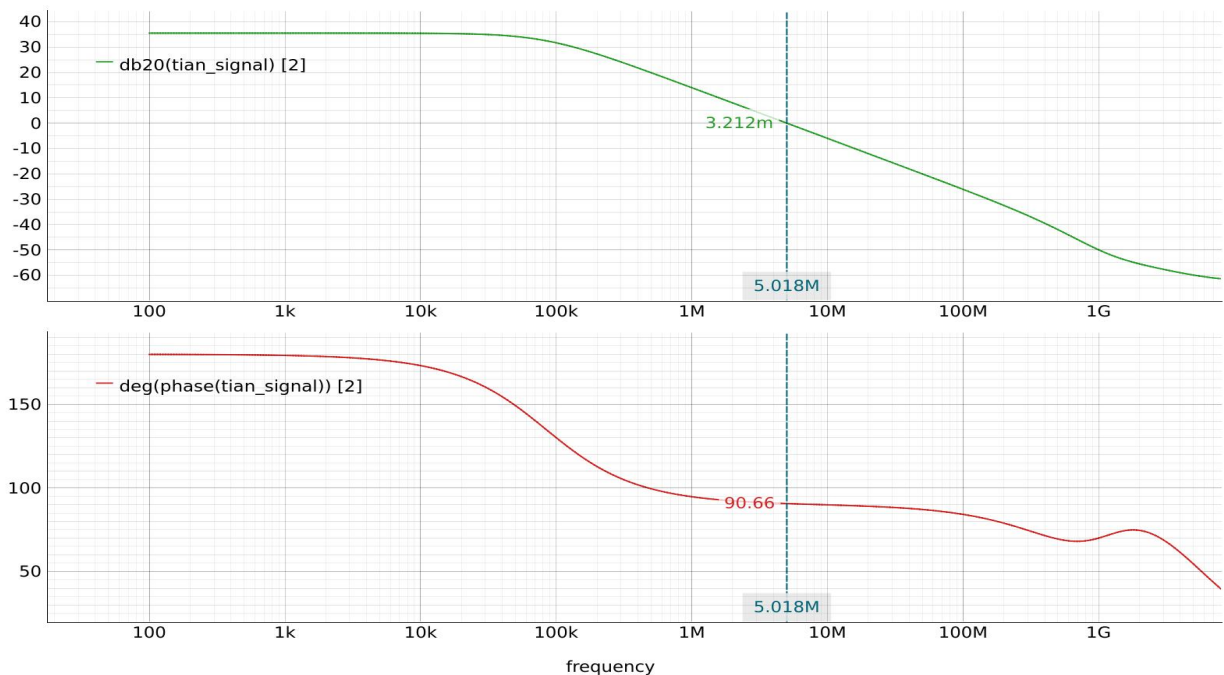
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is the current (and gm) in the input pair exactly equal? Why? Calculate the mismatch?
 No, current not exactly equal M1 has I_D 12.7uA and M2 has I_D 12.2uA
 M1 has 157uS of gm and M2 has 154uS of gm that is because the ota has a finite gain that will make V_{inp} doesn't equal v_{inn} and that will create mismatch between input pair current and transconductance

$$\text{mismatch In current} = \frac{(I_{D1} - I_{D2})}{(I_{D1} + I_{D2})/2} * 100 = 3.8\%$$

$$\text{mismatch In gm} = \frac{(g_{m1} - g_{m2})}{(g_{m1} + g_{m2})/2} * 100 = 1.78\%$$

3.2 Loop Gain (STB Analysis)



Hand calculation

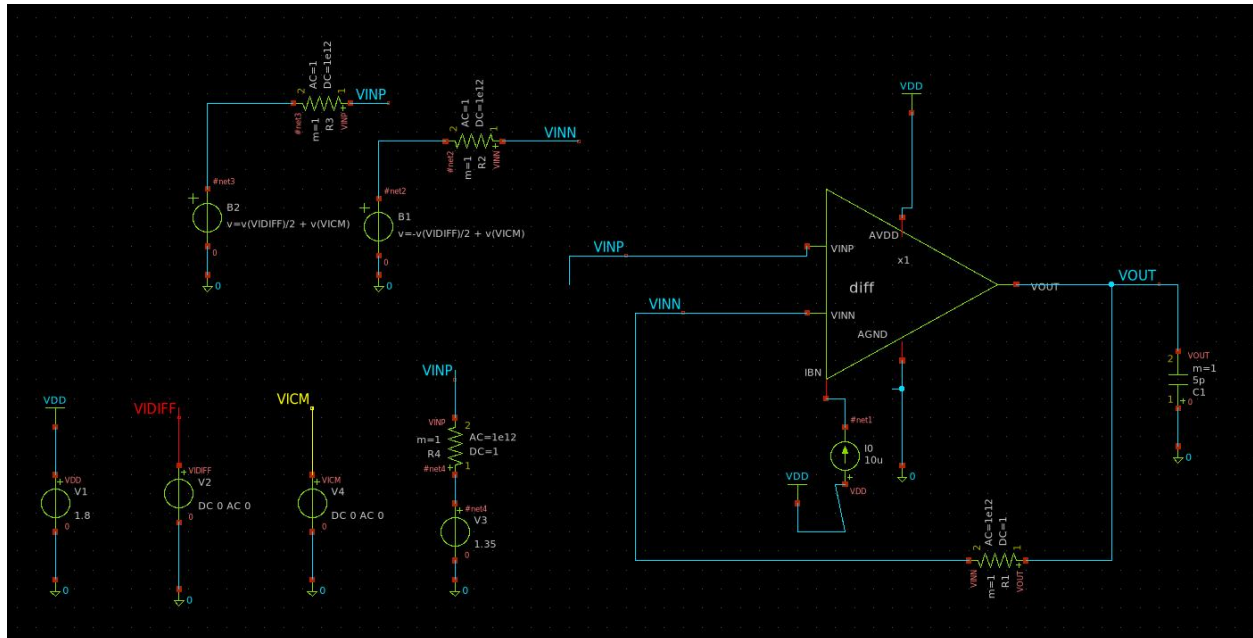
$$A_{v,diff} = g_{m1} \cdot (r_{o1} \parallel r_{o4}) = 54.6091 \text{ (34 dB)} \quad LG \text{ is } = AOL \text{ when } B = 1$$

$$GBW = g_{m1} / (2\pi \cdot CL) = 5.0611 \text{ MHz}$$

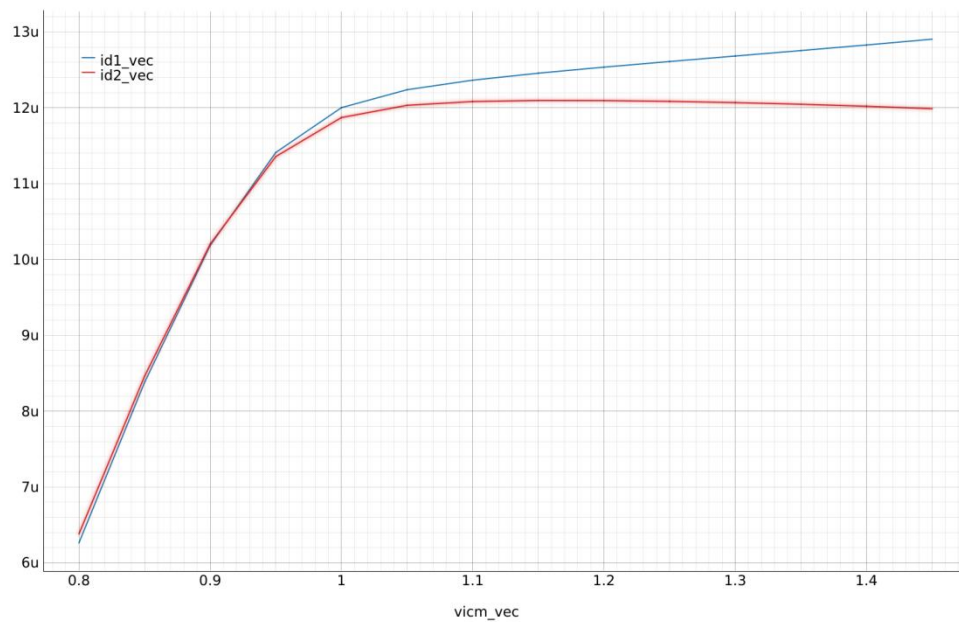
Quantity	Hand Analysis	Open-loop (Part 2)	STB / closed-loop
DC gain (dB)	34 dB	34 dB	35 dB
GBW / crossover freq. (Hz)	5.0611 MHz	5.011 MHz	5.018 MHz

The hand analysis, open loop and stb simulation have approximately same result
 loop gain is equal to BAOL and in this case B is one so LG is exactly AOL

Part 4 — Effect of Mismatch on CMRR

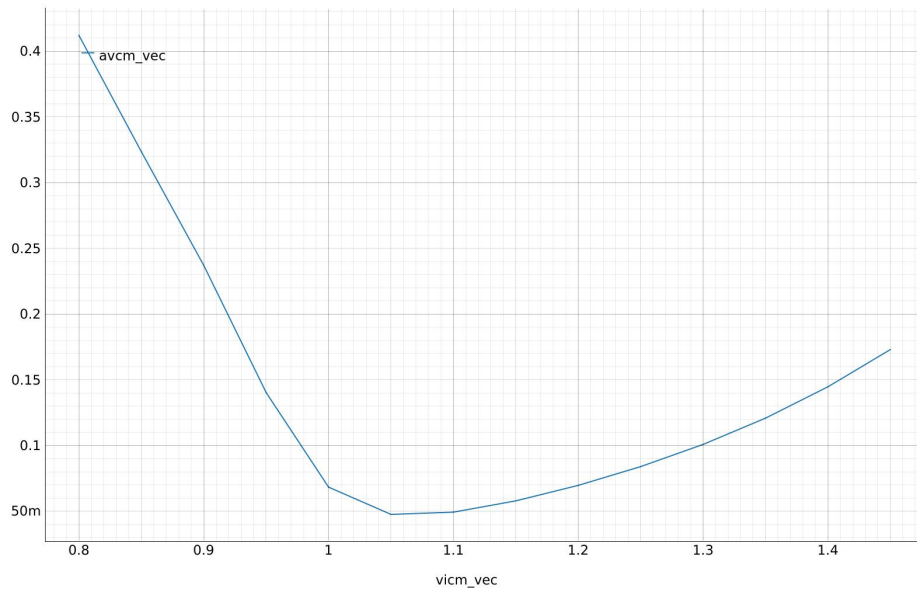


ID(M1) and ID(M2) overlaid vs VICM



this mismatch of tow input pair current is due to the dc feedback , since the designed ota have finite gain so there must be a some difference between two input transistor making their id not equal .

CM gain vs VICM



The graph show that best range of cm is between 1 and 1.15